ROLAND A. KNAPP

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RESEARCH INTERESTS

- Resistance and resilience of aquatic ecosystems to anthropogenic stressors, particularly emerging infectious diseases and nonnative species.
- Impacts of nonnative species on the structure and function of aquatic ecosystems.
- Developing effective strategies to recover endangered amphibians.

EDUCATION

University of California – Santa Barbara: *Ph.D. in Biology* 1992
University of California – Santa Barbara: *B.A. in Aquatic Biology* 1986

ACADEMIC APPOINTMENTS

Earth Research Institute, UC Santa Barbara: *Research Biologist* 2017–present
Marine Science Institute, UCSB: *Research Biologist* 2009–2017
Marine Science Institute, UCSB: *Associate Research Biologist* 2004–2009
Marine Science Institute, UCSB: *Assistant Research Biologist* 1993–2004
Marine Science Institute, UCSB: *Postdoctoral Researcher* 1992–1993

RECENT MEDIA (selected)

- *Science News Explores*, 30 June 2025. Analyze This: Moving frogs to new places helped an endangered species spread
- *Longreads*, 29 May 2025. The fish that climbed a mountain
- *BBC Wildlife Magazine*, 21 November 2024. A long-lost sound has returned to the great mountains of Yosemite
- *USA Today*, 16 November 2024. 'The lakes are alive again': These frogs are back from near extinction
- *Science News*, 08 August 2024. A frog's story of surviving a fungal pandemic offers hope for other species
- *New Scientist*, 10 January 2024. California frog reintroduction is rare victory against fungal pandemic

SELECTED PUBLICATIONS (from 81 peer-reviewed papers; Google Scholar profile)

1. Byrne, A. Q., A. P. Rothstein, R. A. Knapp, and E. B. Rosenblum. 2026. Genomic correlates of disease recovery in natural populations of the Sierra Nevada yellow-legged frog (*Rana sierrae*). Molecular Ecology 36:e70348.
2. Knapp, R. A., M. Q. Wilber, M. B. Joseph, T. C. Smith, and R. L. Grasso. 2024. Reintroduction of resistant frogs facilitates landscape-scale recovery in the presence of a lethal fungal disease. Nature Communications 15:9436.
3. Byrne, A. Q., A. P. Rothstein, L. L. Smith, H. Kania, R. A. Knapp, D. M. Boiano, C. J. Briggs, A. R. Backlin, R. N. Fisher, and E. B. Rosenblum. 2024. Revisiting conservation units for the endangered mountain yellow-legged frog species complex (*Rana muscosa*, *Rana sierrae*) using multiple genomic methods. Conservation Genetics 25:591–606.

4. Knapp, R. A., M. B. Joseph, T. C. Smith, E. E. Hegeman, V. T. Vredenburg, J. E. Erdman Jr, D. M. Boiano, A. J. Jani, and C. J. Briggs. 2022. Effectiveness of antifungal treatments during chytridiomycosis epizootics in populations of an endangered frog. PeerJ 10:e12712.
5. Wilber, M. Q., R. A. Knapp, T. C. Smith, and C. J. Briggs. 2022. Host density has limited effects on pathogen invasion, disease-induced declines and within-host infection dynamics across a landscape of disease. Journal of Animal Ecology 91:2451–2464.
6. Knapp, R. A., G. M. Fellers, P. M. Kleeman, D. A. W. Miller, V. T. Vredenburg, E. B. Rosenblum, and C. J. Briggs. 2016. Large-scale recovery of an endangered amphibian despite ongoing exposure to multiple stressors. Proceedings of the National Academy of Sciences USA 113:11889–11894.
7. Briggs, C. J., R. A. Knapp, and V. T. Vredenburg. 2010. Enzootic and epizootic dynamics of the chytrid fungal pathogen of amphibians. Proceedings of the National Academy of Sciences, USA 107:9695–9700.
8. Vredenburg, V. T., R. A. Knapp, T. S. Tunstall, and C. J. Briggs. 2010. Dynamics of an emerging disease drive large-scale amphibian population extinctions. Proceedings of the National Academy of Sciences, USA 107:9689–9694.
9. Knapp, R. A., K. R. Matthews, and O. Sarnelle. 2001. Resistance and resilience of alpine lake fauna to fish introductions. Ecological Monographs 71:401–421.
10. Knapp, R. A., and K. R. Matthews. 2000. Non-native fish introductions and the decline of the mountain yellow-legged frog from within protected areas. Conservation Biology 14:428–438.

SELECTED EXTRAMURAL GRANTS

- 2021–2026. National Science Foundation. $12,499,323. *BII: Uncovering mechanisms of amphibian resilience to global change from molecules to landscapes* (lead PI C. Richards-Zawacki; Senior Personnel **R. Knapp**).
- 2007–2012. National Science Foundation. $2,500,000. *Collaborative Research: After the crash: factors allowing host persistence following outbreaks of a highly virulent disease* (lead PI C. Briggs, co-PI **R. Knapp**).
- 2002–2007. National Institutes of Health. $2,484,295. *Amphibian disease dynamics in a fragmented landscape* (lead PI C. Briggs, co-PI **R. Knapp**).
- 2019–2022. California Dept. of Fish and Wildlife. $159,666. *Using frog translocations and reintroductions to recover frogs in the Rocky Basin Lakes* (lead PI **R. Knapp**).
- 2018–2022. National Park Service. $96,602. *Restoring genetic diversity of endangered mountain yellow-legged frogs* (lead PI **R. Knapp**).
- 1996–2000. National Science Foundation. $300,000. *Effects of exotic species eradication on the recovery of ecosystem structure and function* (lead PI **R. Knapp**).

PROFESSIONAL ACTIVITIES

Reviewer: 35+ journals including *Ecology*, *Ecology Letters*, *Conservation Biology*, *PLOS One*, *Science*; grant proposals for NSF (Divisions of Environmental Biology and Integrative Organismal Systems), U.S. Fish and Wildlife Service species status assessments and recovery plans.

Scientific Advisor: Technical Advisory Committee & Technical Team assisting USFWS, USFS, CDFW, and NPS in developing and updating a conservation strategy for the mountain yellow-legged frog (1999–present).

Graduate Committees: Served on Ph.D. and Master's committees at UCSB, UC Davis, UC Riverside, and CSU San Francisco (7 students total).

Honors: UCSB Dissertation Fellowship (1991); Phi Beta Kappa (1986).